Transparency and Explainability in Artificial Intelligence

AI Ethics

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Clever Hans (1894-1915)

Figure: Wilhelm von Osten and Clever Hans

The Clever Hans Effect

Figure: Clever Hans performing in 1904

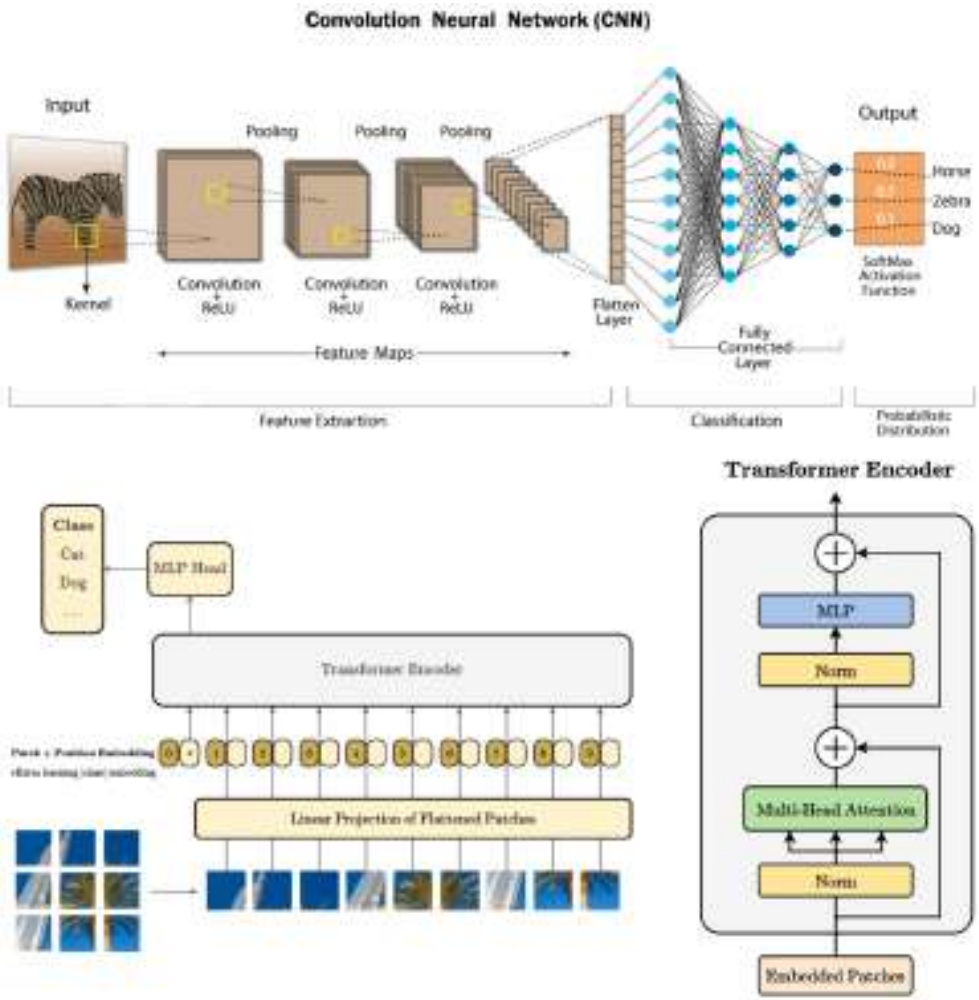
Overview

1. Why Need Explainable AI
2. Explainability Types & Techniques, Interpretability
3. Beyond Explainability: Justifiable Models

Neural Nets Are "Black Boxes"

Why Need Explainable AI

State-of-the-art neural networks are highly complex and lacks transparency.



Neural Nets Can Provably Fail

Why Need Explainable AI

The Pitfalls of Simplicity Bias in Neural Networks

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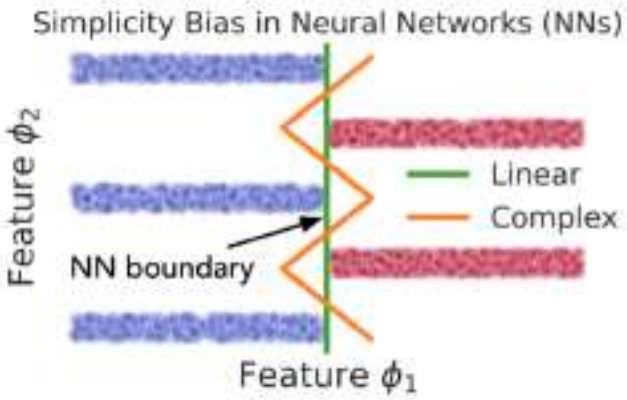
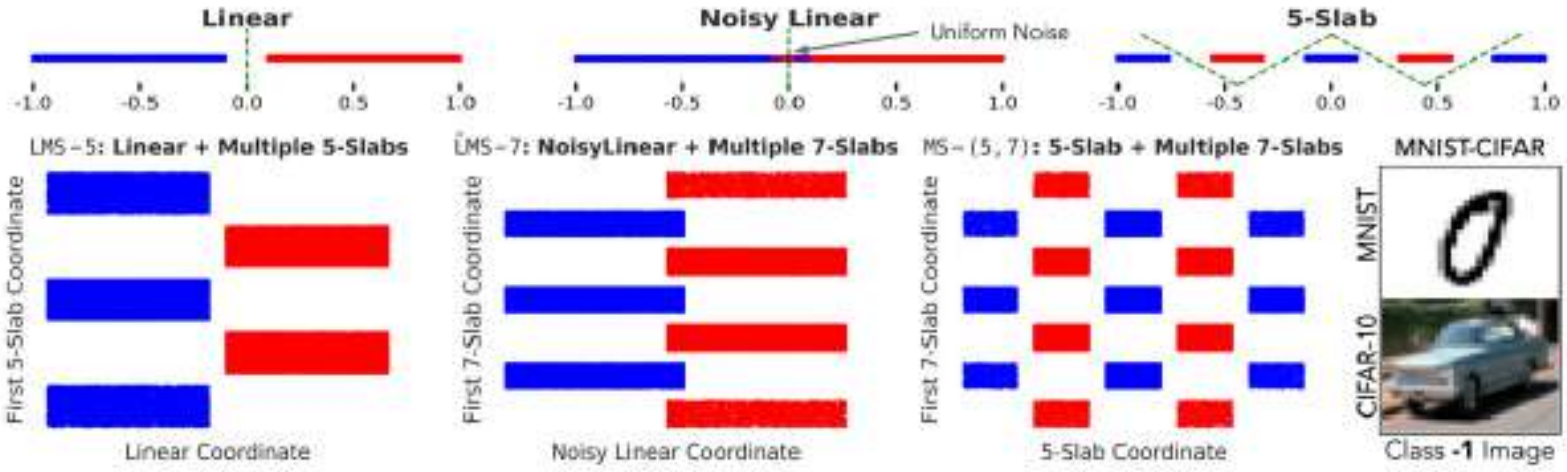
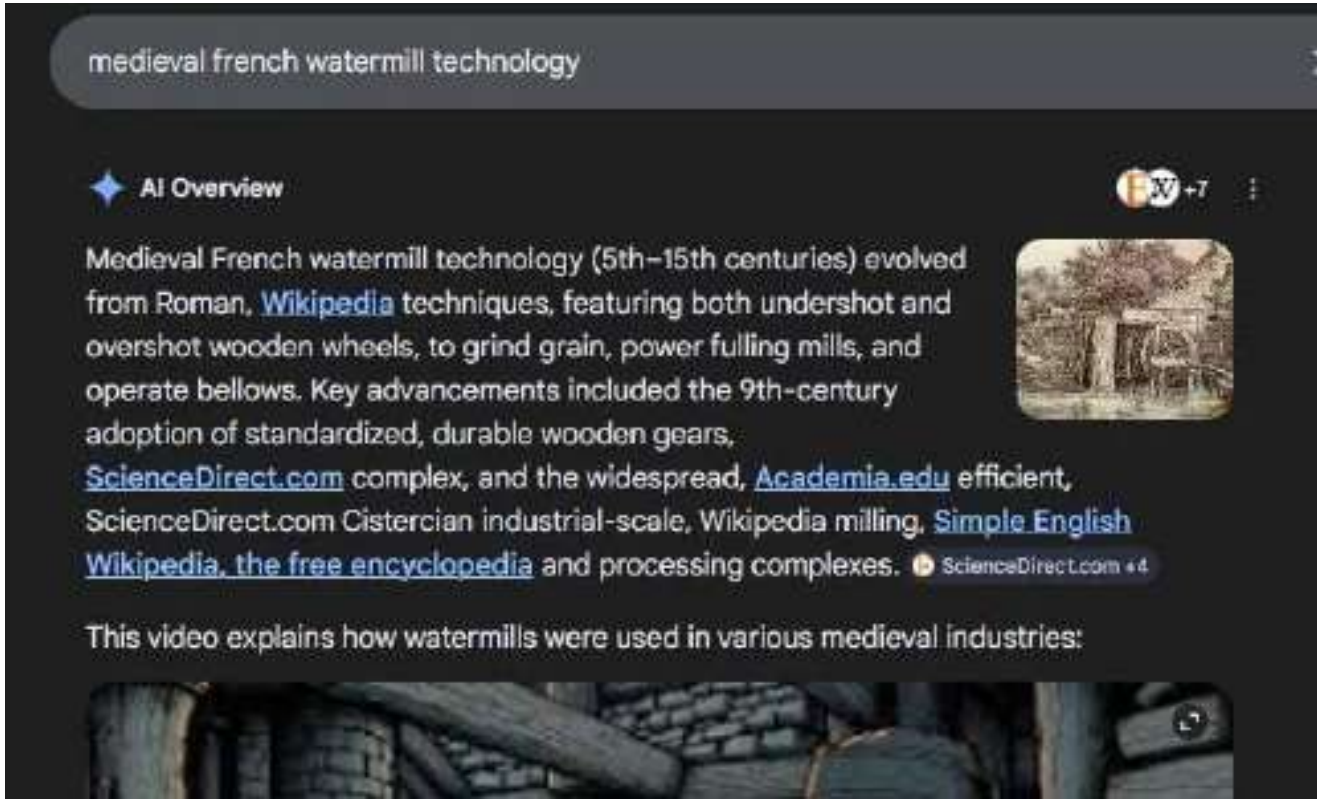




Are AI failures that concerning?



Neural Nets Can Provably Fail: Shortcut Learning

Why Need Explainable AI

Models rely on chest-tubes for pneumonothorax detection.

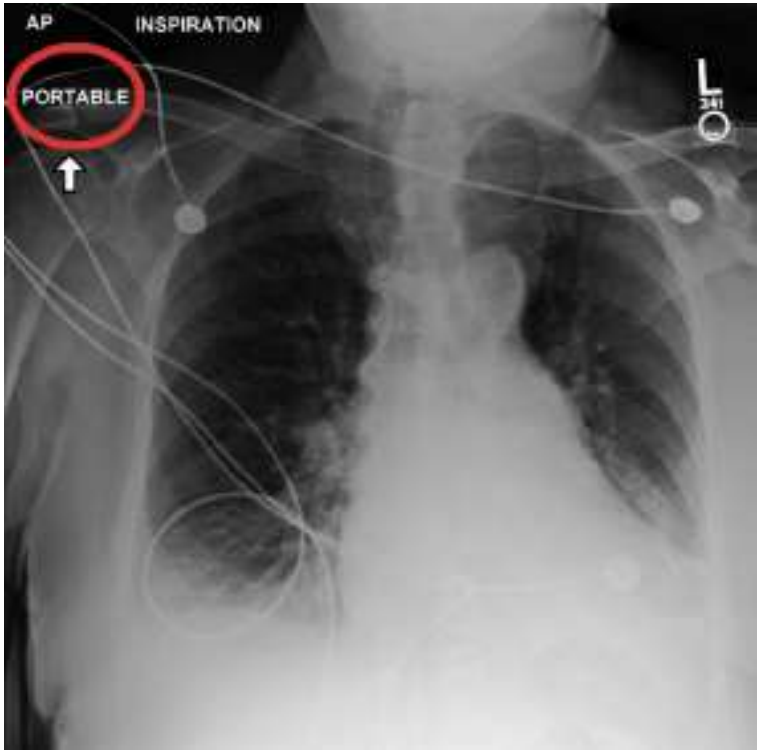


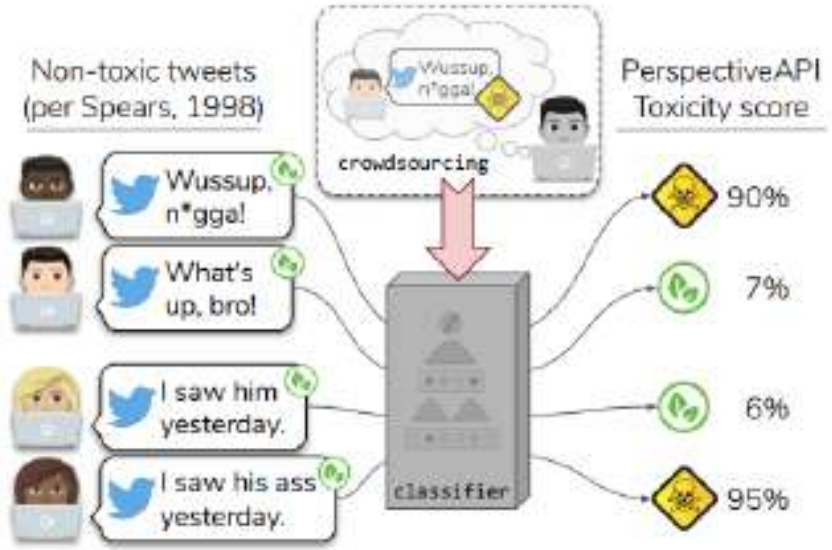
Figure: MIMIC-CRX datasetfootnote[1]

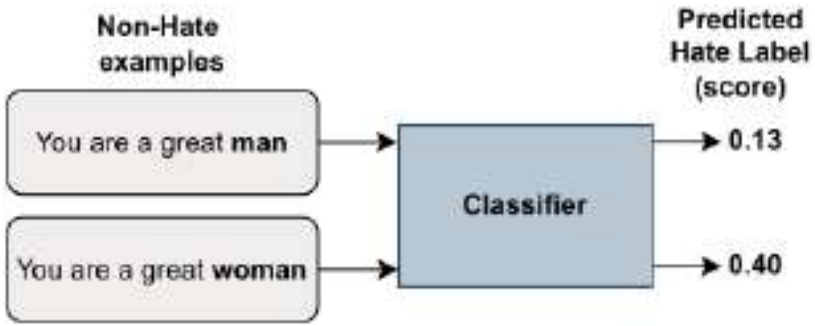
[1] Johnson, A., Pollard, T., Mark, R., Berkowitz, S., & Horng, S. (2024). MIMIC-CXR Database (version 2.1.0). PhysioNet.

Neural Nets Can Provably Fail: Shortcut Learning

Why Need Explainable AI

- Widely used datasets for toxicity / hate-speech detection contains racial [1], gender biases. [2]





[1] Sap et al., The risk of racial bias in hate speech detection, Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, 2019.

[2] Nascimento et al., Gender bias detection on hate speech classification: an analysis at feature-level, Neural Computing and Applications (2025) 37:3887–3905

Lack of Transparency and Ethical Issues

Why Need Explainable AI

Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
Microsoft	94.0%	79.2%	100%	98.3%	20.8%
FACE++	99.3%	65.5%	99.2%	94.0%	33.8%
IBM	88.0%	65.3%	99.7%	92.9%	34.4%

(Click the image to watch on YouTube)

Lack of Transparency and Ethical Issues

Why Need Explainable AI

Black boxes amplify systemic prejudices:

Williams vs. City of Detroit (2020)

- The Incident: Robert Williams was arrested in his driveway in front of his family for a shoplifting crime he did not commit.
- The Black Box Failure:
  - Police used facial recognition on a grainy surveillance photo.
  - The model provided a "possible match" based on an old driver's license.
- The Trust Gap: Officers exhibited **automation bias**, treating a statistical lead as an objective fact. Williams famously noted: *"I hope you don't think all Black men look alike."*

Lack of Transparency and Ethical Issues

Black boxes amplify systemic prejudices:

Alvi Choudhury (Early 2026)

- The Incident: A software engineer was arrested for a burglary 100 miles away; he was held for 10 hours, risking his career and security clearance.
- The Black Box Failure:
  - The algorithm matched him to a suspect nearly a decade younger with different facial proportions.
  - Choudhury: *"I assumed the officer saw I was a brown person with curly hair... and decided to arrest me."*
- The Accountability Gap: The system provided a prediction without a **explanation** that a human could verify for accuracy.

Gender Bias in Hiring (Amazon Recruiting)

(Click the image to watch on YouTube)

Governance, Compliance, and the Law

Why Need Explainable AI

Governments worldwide are shifting toward mandatory transparency in AI. **EU AI Act (2024/2026)** (gold standard) ensures "Right to Explanation" categorizes AI by risk.

- **Unacceptable Risk (Banned)**: Systems that threaten fundamental rights (e.g., social scoring, predatory behavioral manipulation, or untargeted facial scraping).
- **High-Risk (Strictly Regulated)**: Systems in critical sectors like Healthcare, Law Enforcement, Education, and Employment. These require mandatory XAI, human oversight, and robustness.
- **Limited Risk**: Systems like chatbots or deepfakes. These must follow "transparency" rules (e.g., "This image was AI-generated").
- **Minimal Risk**: Spam filters or video game AI (mostly unregulated).

Explainable AI

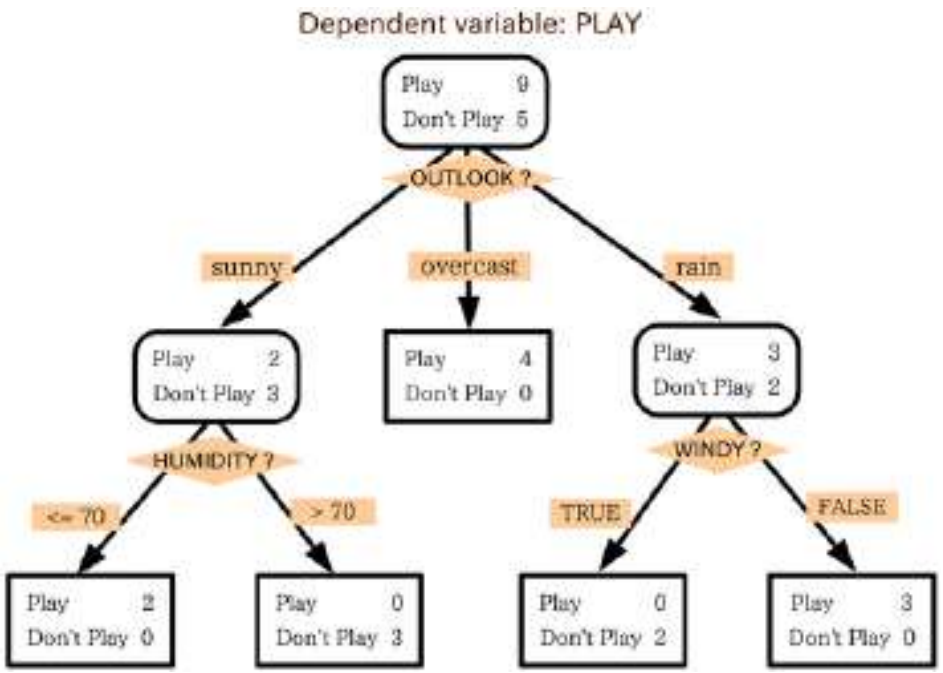
- **Explainability** is the ability to provide clear and understandable explanations for the actions and decisions of an AI system.

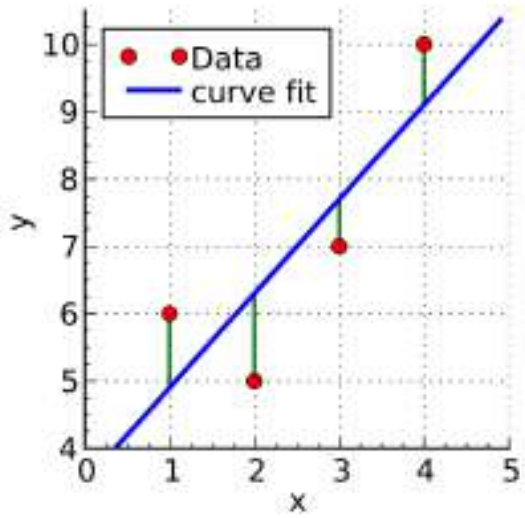
- **XAI system** refer to AI systems that are able to provide explanations for their decisions or predictions and give insight into their behaviors. In short, XAI attempts to understand "WHY did the AI system do X?".

Interpretability vs Explainability

How vs. Why

- **Interpretability:** How easy it is for humans to understand the working principle of an algorithm.
- A model that is highly interpretable is one that is simple and transparent, and whose behavior can be easily understood and explained by humans.

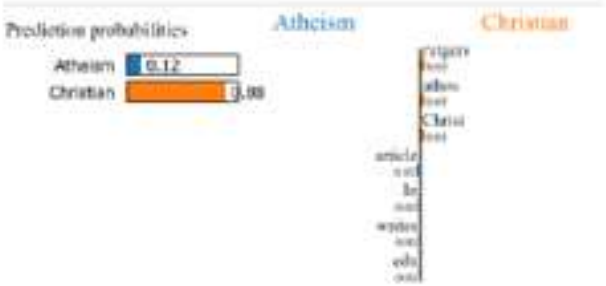


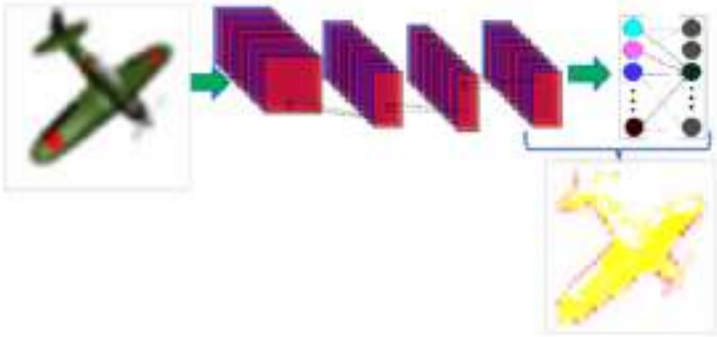


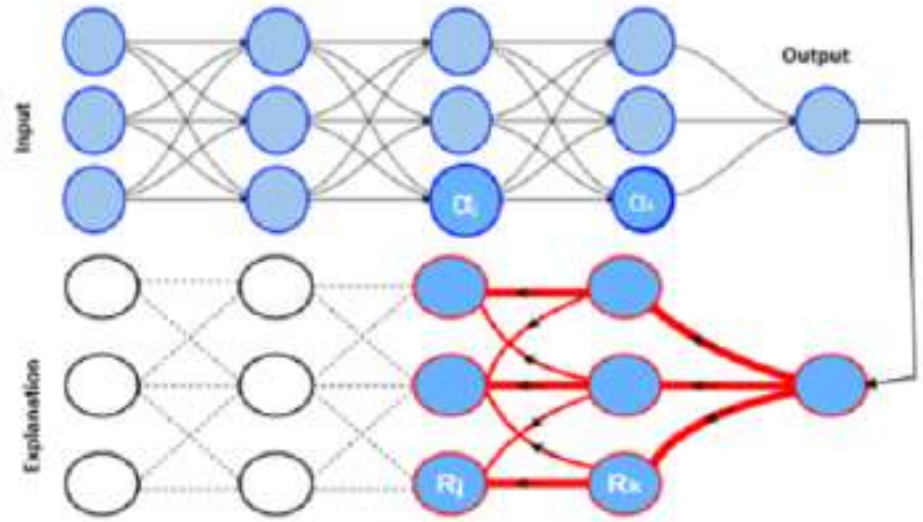
Interpretability vs Explainability

How vs. Why

- **Explainability:** Techniques that opens 'black box' models / algorithms to make it's rationale understandable to human users.







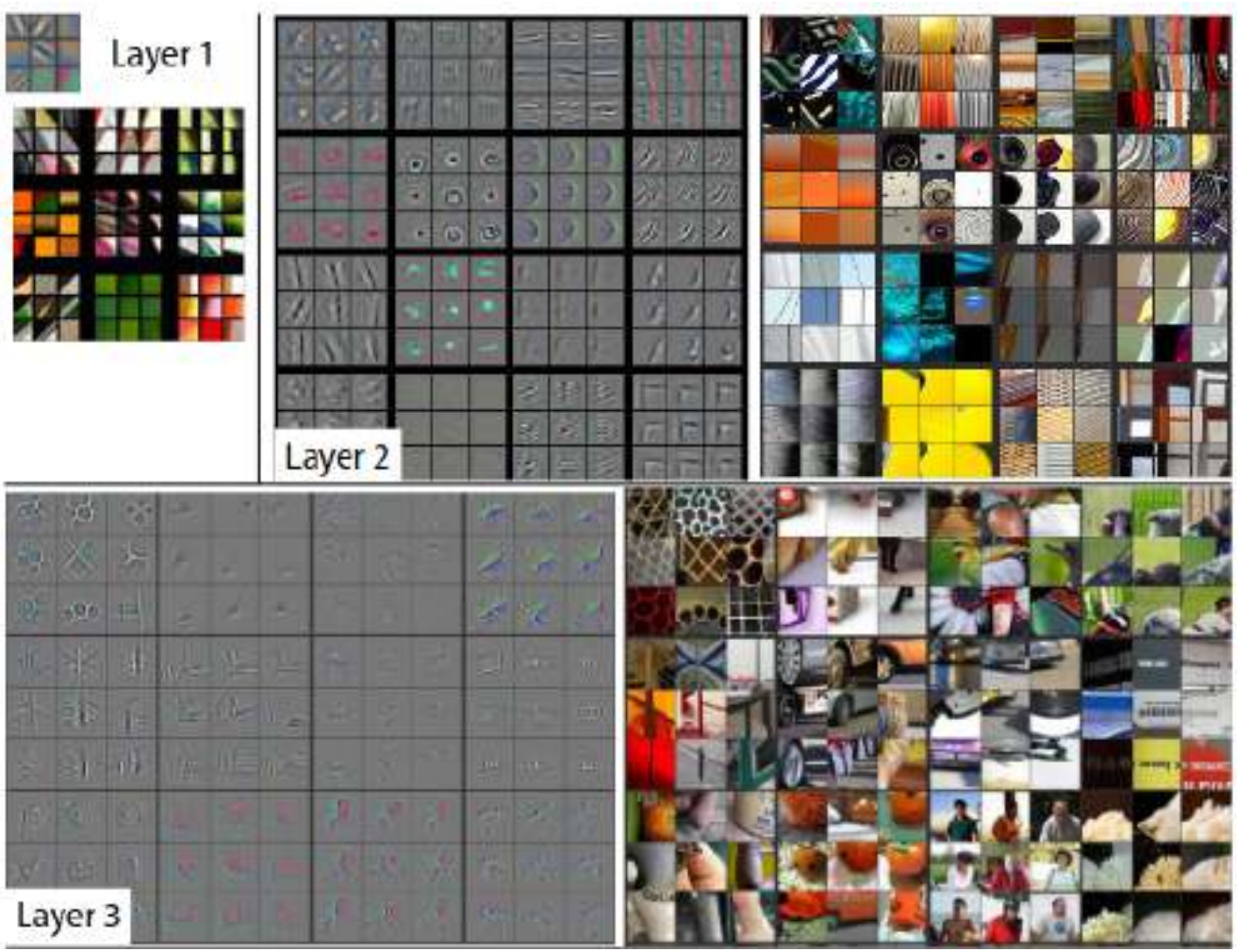
Global vs Local, Post-Hoc vs Ante-Hoc, Model-Specific vs Model-Agnostic

XAI Types

- **Global** techniques provide either an overview or a complete description of the model.
- **Local** explanations are focused on explaining predictions or decisions made by a specific instance or input to a model.
- **Ante-hoc** XAI techniques are employed during the training and development stages of an AI system.
- **Post-hoc** explanation techniques are employed after the AI models have been trained and deployed.
- **Model-Specific** techniques provide explanations for the predictions or decisions made by a specific AI model.
- **Model-Agnostic** techniques can be applied to all AI models and provide explanations of the models.

Deconvolution for CNNs[1]

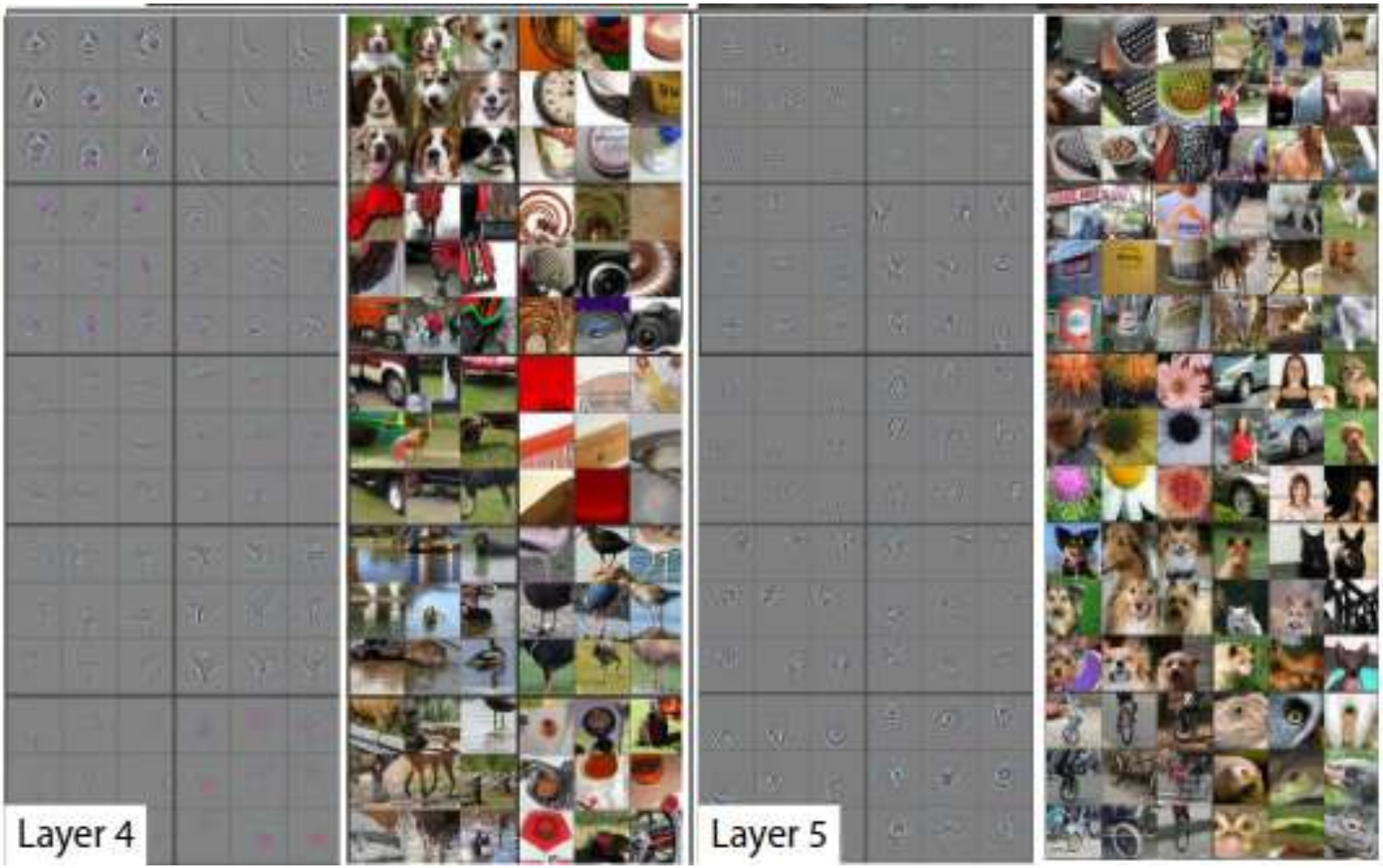
Global, Post-Hoc and Model-Specific



[1] Zeiler, M.D., Fergus, R. (2014). Visualizing and Understanding Convolutional Networks, ECCV 2014

Deconvolution for CNNs[1]

Global, Post-Hoc and Model-Specific



[1] Zeiler, M.D., Fergus, R. (2014). Visualizing and Understanding Convolutional Networks, ECCV 2014

SHAP (SHapley Additive exPlanations)[1]

Local, Post-Hoc and Model-Agnostic

Definition: A game-theoretic approach to explain the output of any machine learning model by attributing the prediction to individual features.

The Shapley Value Formula

The contribution ϕ_i of feature i is the weighted average of its marginal contributions:

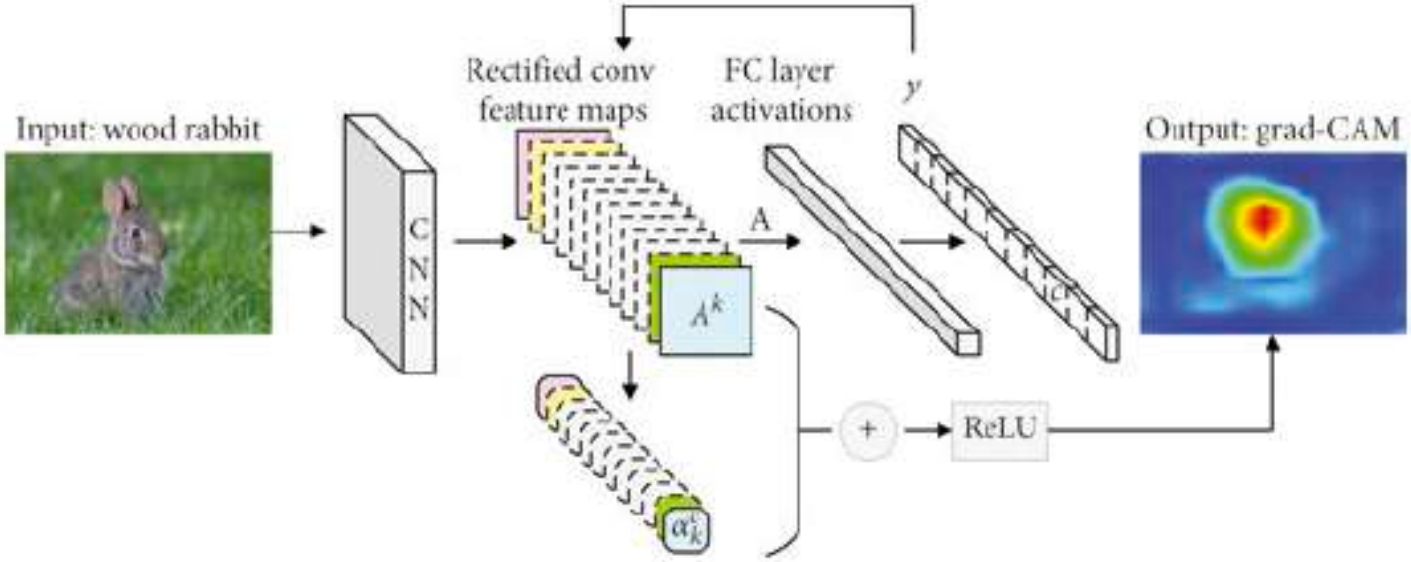
$$\phi_i(f, x) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(M - |S| - 1)!}{M!} [f_x(S \cup \{i\}) - f_x(S)]$$

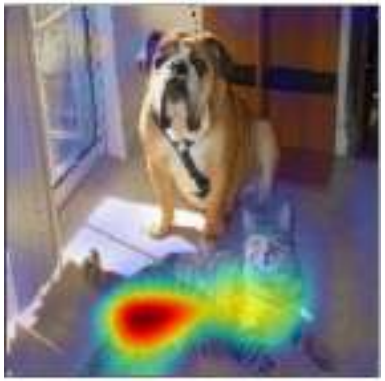
- M: Total number of features ($|N|$).
- S: A subset (coalition) of features excluding feature i.
- $f_x(S)$: The model's expected prediction given features in S.

[1] Lundberg et al., A Unified Approach to Interpreting Model Predictions, NeurIPS '17

Grad-CAM[1]

Local, Post-Hoc and Model-Specific





[1]Selvaraju et al., Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization, ICCV '17

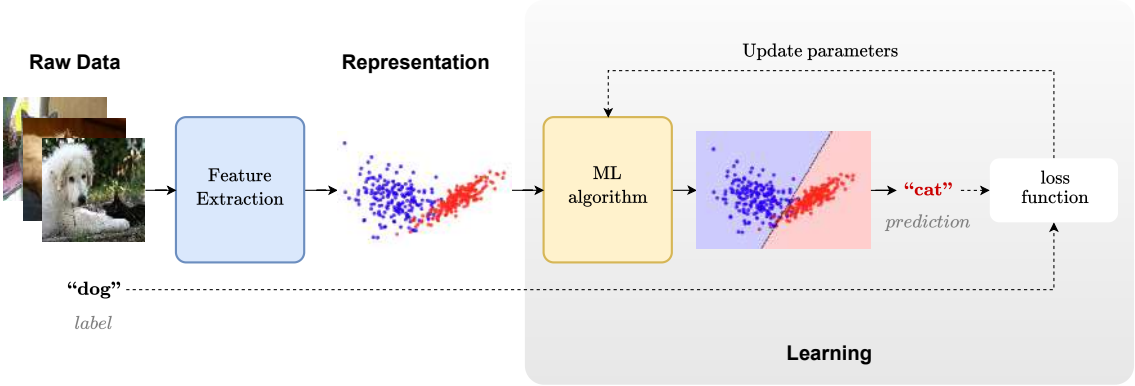
Self-Attention[1]

Local, Ante-Hoc and Model-Specific

[1] Vaswani et al., Attention Is All You Need, NeurIPS '17

Machine Learning

Relies on pre-defined feature extraction techniques

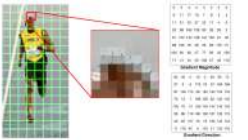


Feature Engineering

Domain-specific features handcrafted by humans

Image

Gradient

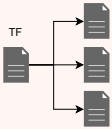


Histogram of Oriented Gradients (HOG)

Text

IDF

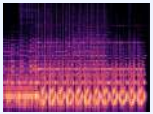
TF



Term Frequency - Inverse Document Frequency (TF-IDF)

Audio / Music

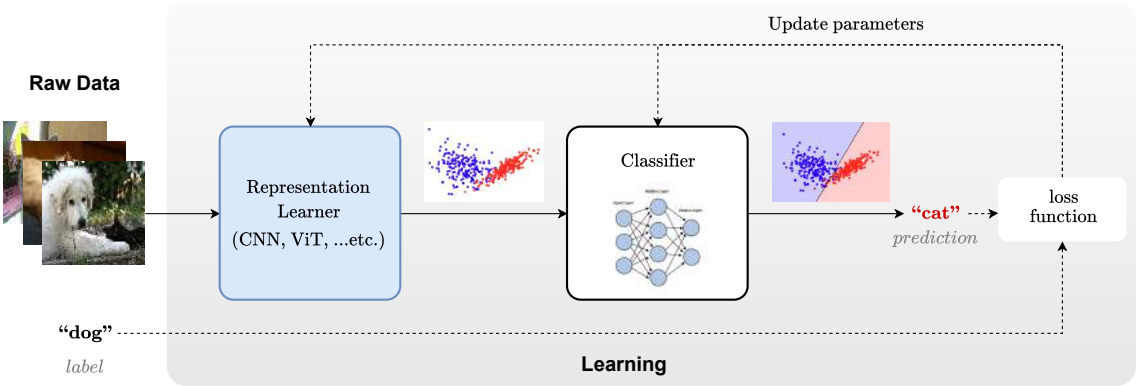
Spectrogram



Mel - Frequency Cepstral Coefficients (MFCC)

Deep Learning

Learned representations



Key Points

- Deep learning **eliminates** manual feature engineering.
- **Human-perspective** is out of the learning process.

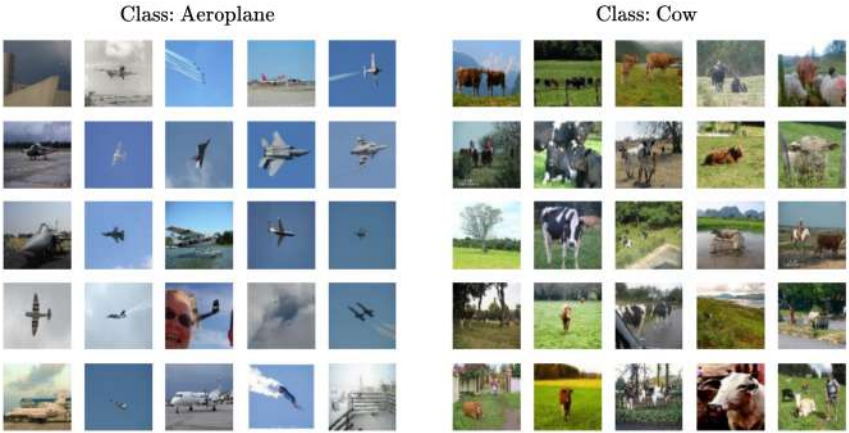
Key Points

- Deep learning **eliminates** manual feature engineering.
- **Human-perspective** is out of the learning process.

- Is it *justified* to rely solely on these automatically extracted feature representations?

Spurious Correlation

Context/background may co-occur with the actual object

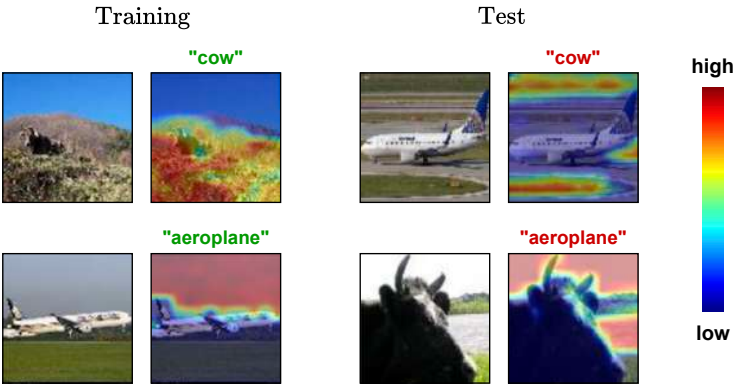


- Classifiers see an image as a whole.
- Context/background takes up majority of the image.

Spurious Correlation

Vision Transformers (ViT) are susceptible to spurious correlations

Classifiers' decisions rely exclusively on the spurious feature and completely avoid relevant features.



Models' decision are not **justified** from a human perspective.

Explainability vs Justifiability

Explainability does not guarantee assured justification

Model

"aeroplane"

neither ***explanable***
nor ***justified***

Model

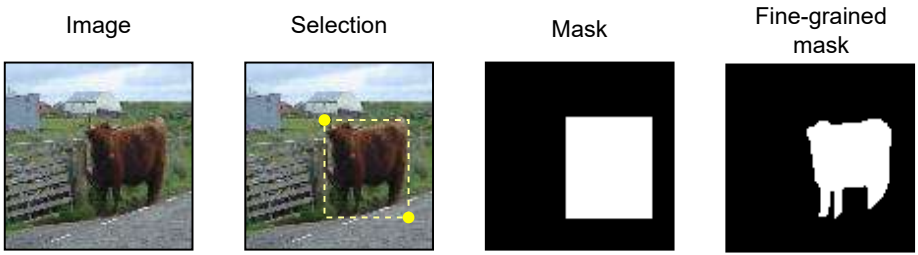
the model predicts
"aeroplane" because
of "sky"

explanable
not ***justified***

Methods

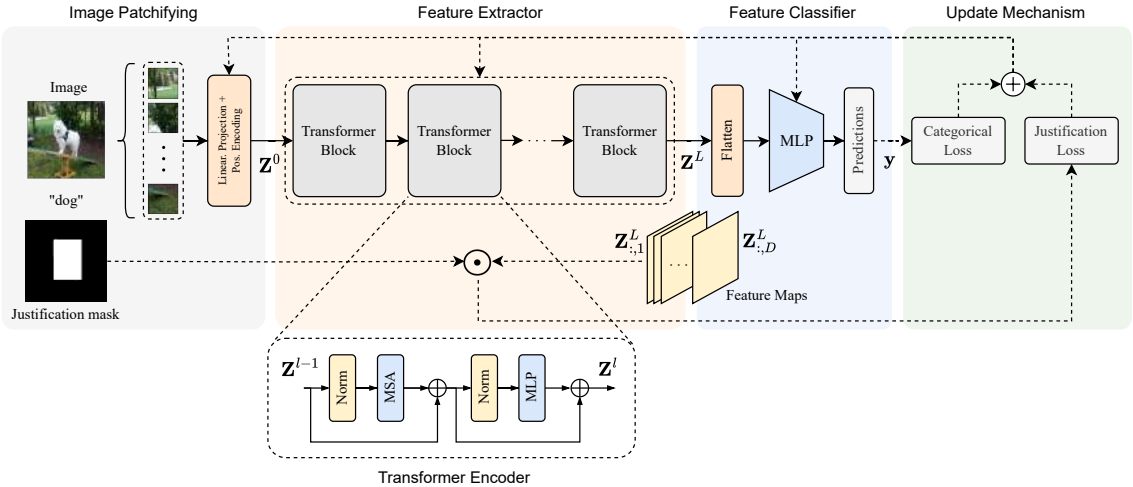
Minimal Additional Guidance

Justification masks to lead ViT towards *justifiable features*



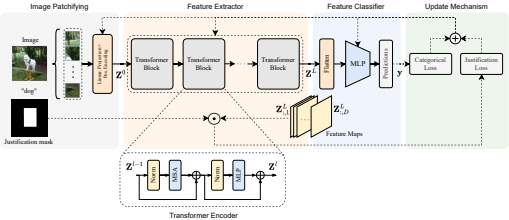
Requires minimal guidance: two clicks per object for a small subset of train set.

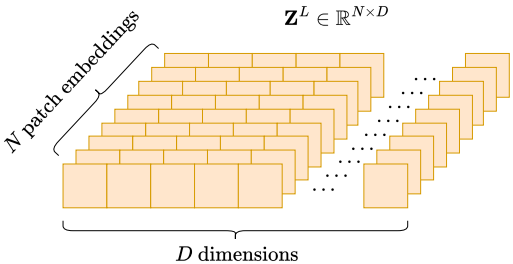
justifiable - Vision Transformer (j-ViT)

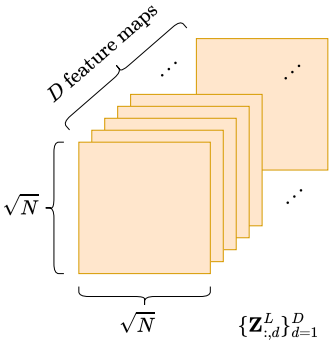


j-ViT

Feature maps







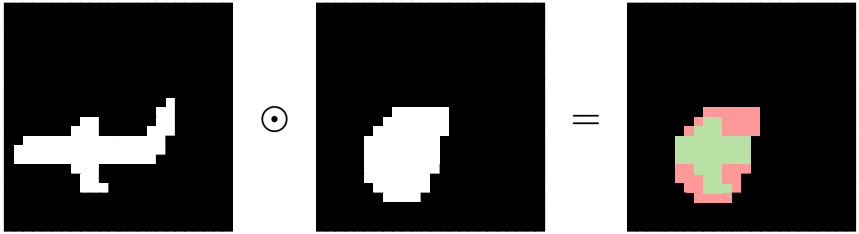
Justification Loss

Proposed loss function

Learning with justification masks: *Justification loss*

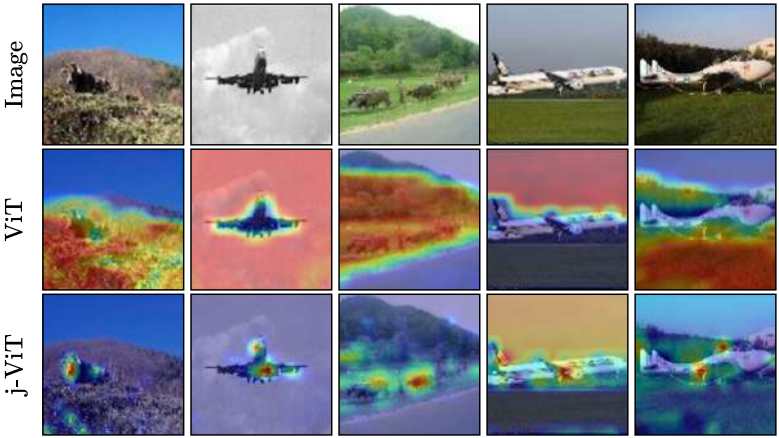
$$\mathcal{L} = \underbrace{-\sum_{i=1}^{C} y_i \log \hat{y}_i}_{\mathcal{L}_{ce}} + \underbrace{\alpha \left(1 - \frac{1}{D} \sum_{d=1}^{D} \left(\frac{\sum_{j=1}^{H} \sum_{k=1}^{W} (\mathbf{A} \odot \mathbf{Z}^L_{:,d})_{j,k}}{\sum_{j=1}^{H} \sum_{k=1}^{W} (\mathbf{Z}^L_{:,d})_{j,k} + \epsilon} \right) \right)}_{\mathcal{L}_{jl}},$$

$\mathbf{A} \in \mathbb{R}^{H \times W}$ is the justification mask.



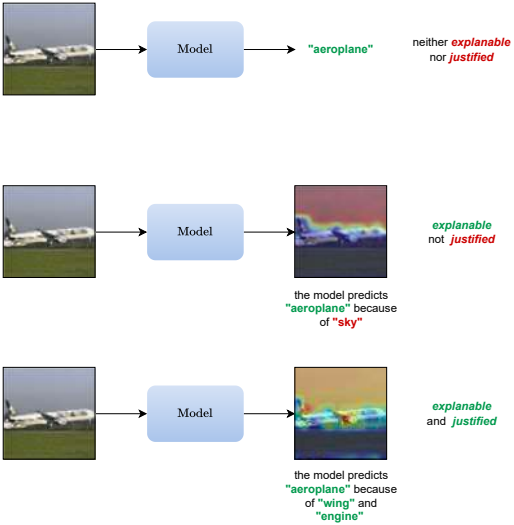
Results

j-ViT is effective in learning justifiable features



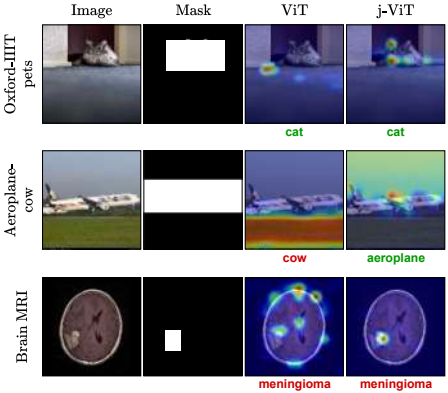
Results

Justification is achieved



Results

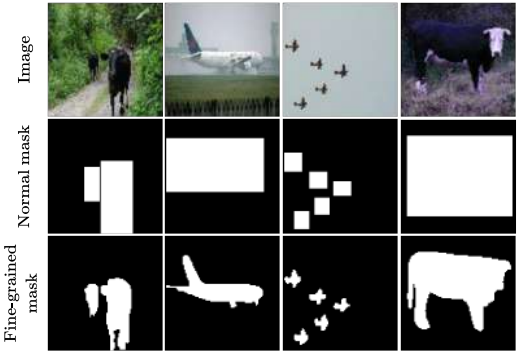
Quantitative and qualitative results



	ViT		jViT	
Dataset	Accuracy (%)	F1-score	Accuracy (%)	F1-score
Oxford-IIIT Pets	66.1 ± 0.13	65.9 ± 0.15	68.0 ± 0.13	67.4 ± 0.12
Aeroplane-Cow	60.0 ± 0.35	60.8 ± 0.36	64.7 ± 0.35	65.3 ± 0.35
Brain MRI	93.0 ± 0.31	93.0 ± 0.30	91.1 ± 0.39	91.1 ± 0.37

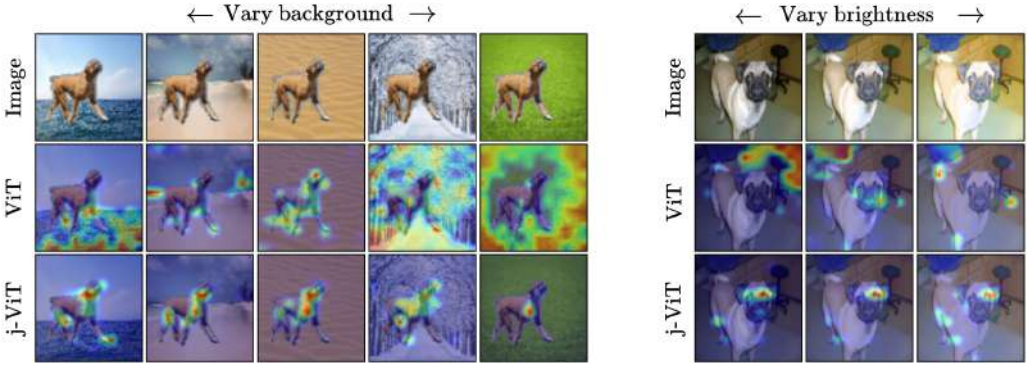
Results

Justification does not require detailed annotations



Results

jViT is robust to contextual variations



Thank You!

Questions?